Avi Mehta

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WORK EXPERIENCE

Truist Charlotte, NC

Data Scientist | Treasury Data Office Dec 2024 - Present

- Led development of data infrastructure and feature pipelines processing 2TB+ of monthly data for a $550B+ banking portfolio, enabling predictive risk models used in regulatory stress testing and enterprise-scale financial workloads
- Engineered General Ledger-to-portfolio reconciliation algorithms into a reusable framework, cutting manual validation effort by 30% and improving high-integrity datasets for model training, evaluation, and downstream analytics
- Partnered directly with risk and compliance stakeholders to convert ambiguous requirements into interactive dashboards, turning complex model output into decisions leadership could act on and improving audit transparency
- Ran statistical trend analysis and anomaly detection across 500K+ loan accounts monthly, surfacing exceptions and edge cases that fed credit risk and early-warning models

Data Scientist | Truist Audit Services June 2023 – Nov 2024

- Developed anomaly detection models for AML ops against noisy transaction data, improving high-risk entity identification by 9% without increasing false-positive rates
- Architected a random sampling framework and orchestrated scalable data pipelines in Python, cutting manual processing by 85%
- Built predictive models forecasting Adjustable-Rate Mortgage portfolio performance, reducing overall risk exposure by 12%
- Accelerated compliance reviews by 30% through a fuzzy-logic inference system that reconciled OFAC and transaction data

Virginia Tech Department of Computer Science Blacksburg, VA

Undergraduate Teaching Assistant | CS3724 Human Computer Interaction Aug 2022 - May 2023

- Held weekly office hours, assisted the professor in writing test and homework material, and graded exams of 150+ students
- Led discussion sections of 48 students weekly, working through problems interactively

Blankly Finance (Backed by Contrary Capital and Rough Draft Ventures) Ann Arbor, MI

Software Engineer Intern Oct 2021 - May 2022

- Engineered scalable user registration flows using React, Flask, and Firebase, with full test coverage using Pytest and Cypress
- Increased Blankly supported exchange integrations by 50% leveraging RESTful APIs and web sockets
- Built and deployed trading bots using the Blankly API, leveraging Docker and AWS EC2, to allow users to simulate trades

The Consulting Group at Virginia Tech Blacksburg, VA

Technology Consultant Oct 2020 - May 2022

- Constructed a cybersecurity system for smart vehicles and conducted A/B testing to evaluate and mitigate vulnerabilities
- Drove M&A due diligence for a fintech merger, emphasizing WealthTech, blockchain, and digital banking markets
- Orchestrated a 5G adoption and utilization roadmap for a healthcare client leveraging scientific + market research

SKILLS

Languages:	Python, Java, JavaScript/TypeScript, SQL, C/C++, SAS
Technologies & Frameworks:	React, Flask, Node, Next.js, Postgres, MongoDB, GraphQL, FastAPI
Libraries:	TensorFlow, Pandas, PyTorch, NumPy, Matplotlib, Scikit
Infrastructure & Tools:	Linux, Git, Docker, Kubernetes, AWS, GCP, Claude Code

PROJECTS (View all on avimehta.info)

Nestor AI

- Launched a no-code agentic AI platform for creating, deploying, and monitoring AI agents with real-time streaming
- Optimized token costs to near-constant via sliding-context windows and async summarization while maintaining output quality
- Designed a reusable agent orchestration layer with multi-provider LLM routing, failover, deterministic fallbacks, and encrypted tenant isolation, supporting 500+ agent runs across varied use cases
- Led 0-1 product development across architecture, UX, deployment, and iteration for scalable rollout

Kernel Mate

- Shipped an LLM-powered terminal assistant as a zero-dependency, cross-platform CLI on PyPI, improving command accuracy from 70% to 95% via context-aware prompt engineering
- Cut response latency from ~2.2s to under 100ms (cached) using SHA-256-based context-aware caching
- Designed a safety-first execution system with risk detection, user confirmation, and automatic undo generation

EDUCATION

Virginia Tech May 2023

B.S. in Computer Science, Cum Laude